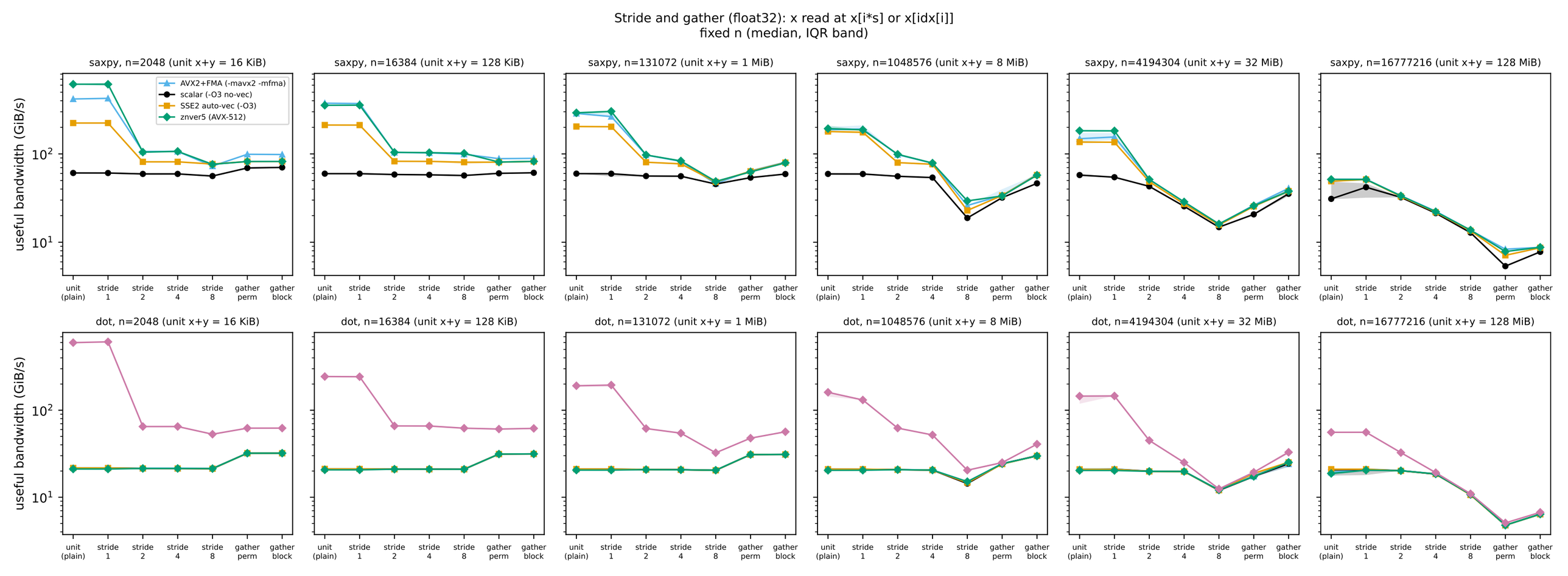
Stride and gather (float32): x read at x[i*s] or x[idx[i]]
fixed n (median, IQR band)
saxpy, n=2048 (unit x+y = 16 KiB)
saxpy, n=16384 (unit x+y = 128 KiB)
saxpy, n=131072 (unit x+y = 1 MiB)
saxpy, n=1048576 (unit x+y = 8 MiB)
saxpy, n=4194304 (unit x+y = 32 MiB)
saxpy, n=16777216 (unit x+y = 128 MiB)
dot, n=2048 (unit x+y = 16 KiB)
dot, n=16384 (unit x+y = 128 KiB)
dot, n=131072 (unit x+y = 1 MiB)
dot, n=1048576 (unit x+y = 8 MiB)
dot, n=4194304 (unit x+y = 32 MiB)
dot, n=16777216 (unit x+y = 128 MiB)
useful bandwidth (GiB/s)
$10^2$
$10^1$
AVX2+FMA (-mavx2 -mfma)
scalar (-O3 no-vec)
SSE2 auto-vec (-O3)
znver5 (AVX-512)
unit (plain)
stride 1
stride 2
stride 4
stride 8
gather perm
gather block